

AYESHKANT RAY

+91-9936666081 | ayeshkantray@gmail.com | [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

EDUCATION

- Vellore Institute of Technology** 2022 - 2026
B. Tech Computer Science & Engineering | CGPA: 8.5
- Kendriya Vidyalaya Sector 8 RK Puram Delhi** June, 2022
Higher Secondary | Percentage: 86.6

SKILLS

- Programming Languages:** Python, C++, SQL
- Libraries & Frameworks:** NumPy, Pandas, Seaborn, Scikit-learn, Matplotlib, Dialogflow CX
- Databases:** MySQL, MongoDB, Database Management System (DBMS), Relational Database Design
- Tools & Platforms:** Git, GitHub, Visual Studio Code, Eclipse, MATLAB, MS Excel, Power BI, Canva
- Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Computer Networks, Operating Systems

EXPERIENCE

- Software Intern – Tonbo Imaging** Nov, 2024 – Jan, 2025
 - Conducted an in-depth analysis of kernel modules and images to identify dependencies and bottlenecks, strategically disabling non-essential components to reduce boot time from 35 seconds to 28 seconds and enhance system performance.
 - Contributed to the **"Kernel Optimization for Faster Boot & Power Efficiency"** project, focusing on reducing system startup time and improving overall energy management.
- Backend Development Intern — StayChat AI** Nov, 2025 – May 2026
 - Integrated frontend and backend reservation APIs to design and manage complete hotel data in MongoDB Atlas** (basic info, rooms & rate plans, rules & policies, media), ensuring real-time consistency across booking workflows.
 - Designed and optimized AI agents using **Dialogflow CX** for automated hotel booking support, improving customer interaction workflows and reducing manual intervention.

PROJECTS

- Library Management System using Python & MySQL** May 2023 – May 2026
 - Engineered a command-line Library Management System with 7 functions across book management, user registration, borrowing, and returns — backed by MySQL across 4 relational tables with full CRUD via mysql.connector.
 - Implemented a 3-tier overdue fine system with a 14-day borrow window, 2-day grace period, and ₹5/day fine calculation; full borrowing history preserved in DB via return date tracking for analytics continuity.
 - Built a live Power BI dashboard connected to MySQL — visualizing genre distribution, monthly borrowing trends, user activity, and a "Wall of Shame" overdue tracker.
 - Hardened the codebase iteratively: parameterized all SQL queries (SQL injection prevention), moved credentials to .env, wrapped all transactions in try/except with rollback, and replaced recursive calls with loop-based flow to eliminate stack overflow risk.
- Rossmann Store Sales Prediction (Python, Pandas, Scikit-learn, Power BI)** April 2026
 - Built an end-to-end sales forecasting pipeline on 1M+ records across 1,115 German drug stores — merging transactional and store metadata, handling domain-aware null imputation, and engineering temporal features (Year, Month, WeekOfYear).
 - Applied time-based 80/20 train-test split to prevent data leakage; used One-Hot Encoding over Label Encoding to preserve nominal category integrity.
 - Trained a Linear Regression baseline: $R^2 = 0.226$, RMSE = 2,700, MAE = 1,960, MAPE = 27.51%; diagnosed heteroscedasticity via residual plots, motivating XGBoost as next iteration.
 - Built a 4-page interactive Power BI dashboard (Sales Overview, Sales Drivers, Store Performance, Model Evaluation) on exported model predictions — covering seasonal patterns, promo impact analysis, and actual vs predicted trends.

CO-CURRICULAR ACTIVITIES

ACHIEVEMENTS

- Completed the 100 Days of Code Challenge, focusing on C++ and Data Structures & Algorithms, and solved 150+ coding problems, significantly strengthening problem-solving and algorithmic thinking skills.
- Co-authored and secured a Government of India registered copyright for *"Cool Veil"*, a smart thermoelectric cooling jacket; collaborated with a multi-disciplinary engineering team to develop an innovative wearable technology solution.
- Achieved 2nd Runner-up at the SGU Hackathon, themed on Mental Health AI, for developing a culturally-aware AI-based screening platform.

ACTIVITIES

- DSW Office Member, VIT Bhopal University (Dec 2022 – Aug 2025): Contributed to planning and execution of university events, workshops, and extra-curricular activities; served as Managing Editor for the monthly newsletter (Apr 2023 – Jun 2023).
- Event Organizer, EV Club – VIT Bhopal University: Organized and managed major technical and non-technical events; led coordination efforts and supported partnership initiatives to enhance club operations.

CERTIFICATIONS

- Amazon Web Services** - Introduction to Machine Learning on AWS June 2023
- Johns Hopkins University** - A Crash Course in Data Science June 2023

ADDITIONAL

- Hobbies**- Reading, Music, Gaming, Swimming.